

Adaptive Speculation Length Optimization for Speculative Decoding in Large Language Model Inference

Pratham Ramachandra Bharati

M.S. Applied Machine Learning, University of Maryland, College Park

pratham.bharati03@gmail.com

Code: github.com/prathambharati/msml604-speculative-decoding

Abstract

Speculative decoding accelerates inference in large language models by using a small draft model to propose multiple tokens that a larger target model verifies in a single forward pass. The wall-clock speedup depends on a single hyperparameter, the speculation length k , which is the number of tokens drafted per round. Choosing k too small under-utilizes the draft; choosing k too large wastes draft work on tokens that get rejected. We frame the choice of k as a constrained optimization problem that minimizes expected per-token wall-clock time, subject to an empirically fitted latency model and the integrality constraint $k \in \mathbb{Z}^+$. Using 425 measured runs of speculative decoding (Qwen 2.5 0.5B as draft, Qwen 2.5 7B as target, NVIDIA A100 80GB) we fit the latency model with $R^2 = 0.9999$ and check convexity of the objective in both k and $\log k$. We then compare three optimization methods on the same data: convex relaxation, multi-armed bandit (UCB1 and Thompson sampling), and Bayesian optimization with a Gaussian-process surrogate. Three of the four solvers identify the empirical optimum $k^* = 3$. The fourth selects $k = 5$, which we attribute to surrogate noise from limited sampling. Our second contribution is theoretical: statistical testing rejects the IID acceptance assumption used in every prior speculative decoding analysis ($p < 0.0001$ for all $k \geq 4$), with per-position acceptance rate rising from 0.74 at the first position to 0.87 at later positions. We derive a closed-form non-IID extension of the cost model under linear position-dependent acceptance, give the first-order optimality condition, and show that the new model reduces total absolute prediction error of expected tokens per round by 78.8 percent on the full dataset and 23.1 percent under leave-one-category-out cross-validation.

Keywords: speculative decoding, convex optimization, multi-armed bandits, Bayesian optimization, large language models, hyperparameter selection.

I. INTRODUCTION

Inference in modern large language models is dominated at deployment time by the autoregressive decoding loop. Each new token requires a full forward pass through a network with billions of parameters. On NVIDIA A100 and H100 accelerators this loop is severely memory-bound at batch size one. The GPU spends most of its time reading weights from high-bandwidth memory rather than performing arithmetic, and compute utilization in this regime is typically below one percent of peak FLOPS [1], [2]. The bottleneck is fundamental and persists regardless of how powerful the underlying compute units become, since the rate at which weights can be streamed from memory is fixed by the hardware specification of the GPU.

Speculative decoding [3], [4] addresses this bottleneck by splitting work between two models. A small draft model proposes k candidate tokens autoregressively, and the larger target model verifies all k in a single batched forward pass. A token is accepted whenever the target's distribution agrees with the draft's; the procedure is mathematically equivalent to sampling from the target. When draft and target agree often, a single round of verification commits multiple tokens for the cost of approximately one target forward pass. The draft model is typically several orders of magnitude smaller than the target, which means each draft step is much faster than each target step, and the verification cost is amortized over multiple committed tokens.

The realized speedup depends sharply on the speculation length k , a hyperparameter chosen by the practitioner. Choosing k too small forgoes speedup that the draft could provide; choosing k too large wastes draft computation on tokens that are later rejected. The trade-off is governed by the acceptance rate of the draft and by the relative speeds of the two models. The standard cost model in [3] assumes per-position acceptance is independent and identically

distributed across positions within a speculation round, with a single acceptance probability α . Under this assumption the optimal k is well defined and analytically tractable. The IID assumption itself, however, has not been empirically tested at the level of granularity needed to confirm or reject it. Whether it holds in practice, and whether the analytical optimum it predicts matches what is actually observed, are open empirical questions.

This paper makes three contributions. First, we frame the choice of k as a discrete optimization problem with an explicit objective derived from an empirically fitted latency model ($R^2 = 0.9999$) and show that the objective is convex on the operating range, both in k directly and under the substitution $u = \log k$. Second, we compare three optimization methods on the empirical objective: convex relaxation through CVXPY-style continuous grid search, multi-armed bandit through UCB1 and Thompson sampling, and Bayesian optimization through a Gaussian-process surrogate. We characterize their sample complexity and accuracy. Third, we empirically refute the IID acceptance assumption ($p < 0.0001$ in all tests we ran), derive a closed-form non-IID extension of the speculative decoding cost model under linear position-dependent acceptance, and validate that the new model reduces total absolute prediction error of expected tokens per round by 78.8 percent on the full dataset and 23.1 percent under leave-one-category-out cross-validation.

The remainder of this paper is organized as follows. Section II reviews related work. Section III sets up the optimization problem precisely. Section IV analyzes convexity. Section V describes the three optimization methods. Section VI introduces the non-IID theoretical extension. Section VII details the experimental setup. Section VIII presents results. Section IX discusses implications and limitations. Section X concludes.

II. RELATED WORK

A. Speculative decoding

Speculative sampling was introduced by Leviathan, Kalman, and Matias [3] at Google and concurrently by Chen et al. [4] at DeepMind. Both analyses rely on a single per-position acceptance probability α and derive expected speedup as a function of α and k under the IID assumption. An earlier blockwise parallel decoding scheme by Stern, Shazeer, and Uszkoreit [5] prefigured the modern formulation. More recent work has extended speculative decoding to tree-structured drafts (Medusa [6], EAGLE [7]) and to self-speculation without a separate draft model (LayerSkip [8], Lookahead Decoding [9]). The hyperparameter k is treated in most of this prior work as a fixed constant, often four or five, or selected by ad hoc grid search. We are not aware of prior work that frames the choice of k as a constrained optimization problem and compares principled solvers, nor of prior work that empirically tests the IID acceptance assumption at per-position granularity.

B. Adaptive and online speculation

Several recent works observe that the optimal speculation length depends on the prompt and propose adaptive schemes. SpecInfer [10] and BiLD [11] adjust draft behavior dynamically based on observed rejection patterns, and SpecBench [12] provides a benchmark suite. None of these works derive an explicit cost model that is then solved by a generic optimization method, and none challenge the IID acceptance assumption that underlies the standard analysis.

C. Convex optimization, bandits, and Bayesian optimization

The three solvers we compare draw on three established literatures. Convex optimization methods such as gradient descent, Newton's method, and interior-point are surveyed in Boyd and Vandenberghe [13]. CVXPY [14] provides a disciplined modeling framework for convex programs. Multi-armed bandit methods using upper-confidence bounds (UCB1 [15]) and posterior sampling (Thompson sampling [16]) are well suited to discrete arms with stochastic rewards, and they are widely used in online learning when the reward distribution is unknown. Bayesian optimization with Gaussian-process surrogates is reviewed in Frazier [17]; we use the BoTorch implementation [18]. We do not claim novelty in any of these methods individually. Our contribution is to formulate the speculation-length problem as a clean instance to which all three apply, and to report their relative performance under matched experimental conditions.

III. PROBLEM FORMULATION

This section sets up the optimization problem precisely. We introduce notation, define the objective function, fit an empirical latency model, state the standard IID acceptance model used in prior work, and combine these into a discrete optimization problem with a continuous relaxation amenable to standard solvers.

A. Notation

We adopt the following notation throughout. Let $k \in \{1, 2, \dots, K_{\max}\}$ denote the speculation length, where $K_{\max}$ is an upper bound determined by memory and the maximum useful round size. We use $K_{\max} = 12$ throughout this paper, since values beyond 10 are uninteresting in our experiments. Let t_d denote the per-token wall-clock time to run one forward pass of the draft model, and let $t_v(k)$ denote the wall-clock time to verify a sequence of k candidate tokens through one batched forward pass of the target model. Let $\alpha(j)$ be the conditional probability that the candidate token at position $j \in \{0, 1, \dots, k-1\}$ is accepted by the target, given that all positions 0 through $j-1$ were accepted. Let $N(k)$ denote the random variable equal to the number of new tokens committed per round, including the one bonus token from the target's distribution at the rejection point or after position k . Throughput is reported in tokens per second and equals the reciprocal of the per-token cost.

B. Objective

The quantity to minimize is expected wall-clock time per generated token. The total time of a single speculation round is

$$T(k) = k \cdot t_d + t_v(k)$$

and the expected number of tokens produced is $E[N(k)]$. The expected per-token cost is therefore the ratio of these two quantities,

$$f(k) = T(k) / E[N(k)]$$

We seek the integer minimizer

$$k^* = \arg \min_{k \in \{1, \dots, K_{\max}\}} f(k)$$

Smaller $f(k)$ corresponds to higher token throughput, which is the practically interesting quantity. Throughput in tokens per second equals the reciprocal of $f(k)$ when f is measured in seconds per token. We report both throughput and per-token latency in our results, since they convey the same information from different angles.

C. Latency model

We empirically fit the round time $T(k)$ as an affine function of k . Across all 400 speculative runs in our dataset, the relationship is described very accurately by

$$T(k) = a \cdot k + b, \quad a = 31.07 \text{ ms}, \quad b = 107.80 \text{ ms}$$

with $R^2 = 0.9999$. The slope a captures the per-token cost of a draft model forward pass plus the marginal cost of an extra token in the target's batched verification. The intercept b captures fixed per-round overhead from kernel launches, KV-cache truncation, and a single target forward pass on the committed bonus token. The high goodness of fit indicates that no second-order term is needed in the operating range we explore. In particular, the linear scaling of verification cost with k holds because the target's batched forward pass on k tokens is dominated by KV-cache reads from memory, which are roughly proportional to k . Section VII describes the experimental conditions, and Section VIII shows the fit visually (Fig. 1).

D. Acceptance model

Standard analyses [3], [4] assume that $\alpha(j) \equiv \alpha$ is constant in j (the IID assumption). Under this assumption the expected number of accepted draft tokens follows a truncated geometric distribution and $E[N(k)]$ admits the closed form

$$E[N(k)]_{\text{IID}} = \frac{1 - \alpha^{k+1}}{1 - \alpha}$$

We will show in Section VI that the IID assumption is empirically violated in our data and propose a position-dependent generalization. For the present section we proceed under the standard assumption to keep the formulation comparable with prior work, and to set up a baseline against which the non-IID extension can be evaluated.

E. Optimization problem

Combining the latency model and the IID acceptance model, the optimization problem becomes

$$k^* = \arg \min_{k \in \{1, \dots, K_{\max}\}} \frac{(a \cdot k + b)(1 - \alpha)}{1 - \alpha^{k+1}}$$

This is a one-dimensional discrete optimization problem with a non-convex feasible set $\{1, 2, \dots, K_{\max}\}$, an objective parameterized by three empirically observable scalars (α, a, b) , and a continuous relaxation that we analyze in the next section. Although the discrete formulation has only $K_{\max} = 12$ candidates and could be solved by exhaustive enumeration, we are interested in characterizing the structure of the problem so that the same machinery applies to richer settings, for example when α depends on prompt features and is itself a learned function.

IV. CONVEXITY ANALYSIS

A natural question, given the form of $f(k)$, is whether the relaxed objective is convex, since convex objectives admit efficient global solvers and certify optimality. We answer this question both directly in k and after a logarithmic change of variable. Both checks confirm convexity on the operating range, which has direct implications for the choice of solver in Section V.

A. Convexity in k

Treating k as a continuous variable, define

$$f(k) = \frac{(a \cdot k + b)(1 - \alpha)}{1 - \alpha^{k+1}}$$

The numerator $(a \cdot k + b)(1 - \alpha)$ is affine in k and therefore convex. The denominator $1 - \alpha^{k+1}$ is monotone increasing in k (assuming $\alpha \in (0, 1)$) and concave, since its second derivative is $-(\log \alpha)^2 \cdot \alpha^{k+1}$, which is negative for $\alpha < 1$. The ratio of an affine function to a positive concave function is not, in general, convex; the structural argument fails. We therefore verify convexity numerically.

We compute the second derivative $f''(k)$ on a fine grid $k \in [1, 12]$ using the empirically estimated parameters $\alpha = 0.815$, $a = 31.07$ ms, $b = 107.80$ ms. Across the entire grid we observe $f''(k) \in [0.078, 16.13]$, strictly positive throughout (Fig. 2). The minimum value of $f''(k)$ is attained near $k = 12$ where the curvature is small but still positive. We conclude that f is convex in k on the operating range. This is sufficient to guarantee that any local minimum of the continuous relaxation is the global minimum, and that the integer optimum is one of the two integers adjacent to the continuous minimizer.

B. Convexity under log substitution

We additionally check convexity under $u = \log k$, an option suggested by the multiplicative form of α^{k+1} in the denominator. Define $g(u) = f(e^u)$. The chain rule gives $g''(u) = e^u \cdot f'(e^u) + e^{2u} \cdot f''(e^u)$, which is positive when f is increasing and convex (which holds for k above the optimum). For k below the optimum the first term is negative but small in magnitude, and we verify numerically that the sum remains positive. Numerical evaluation of d^2g/du^2 on $u \in [0, \log 12]$ yields values in $[0.080, 16.13]$, strictly positive. So g is also convex; the optimization can be expressed in either variable. The log substitution is interesting because it places the candidate values of k on a more natural scale (an additive shift in u corresponds to a multiplicative shift in k), and is the substitution we would naturally use in a Bayesian optimization framework.

C. Implications

Convexity has two practical consequences for our work. First, the continuous relaxation can be solved in closed form by the first-order condition $f'(k) = 0$, which we solve numerically since α^{k+1} makes the equation transcendental. Differentiating f with respect to k and setting the result to zero gives a fixed-point equation that can be solved by

bisection in less than ten iterations to machine precision. Second, comparing the convex relaxation against the multi-armed bandit and Bayesian solvers becomes an apples-to-apples comparison of how well each method exploits this structure. The convex relaxation should be exact (modulo rounding to integer); the bandit should converge to within statistical noise; the Bayesian method should also converge but with higher variance for small sample sizes. We verify these expectations empirically in Section VIII.

V. OPTIMIZATION METHODS

We compare three families of solvers, all operating on the same empirical objective derived from our latency and acceptance models. They differ in how much structural information they assume about the objective, and in how they balance exploration against exploitation when the objective is observed only through noisy samples. Convex relaxation assumes the most structure (continuous convex function) and produces the most accurate answer in the fewest evaluations. Multi-armed bandit assumes the least structure (only that the objective has a finite number of discrete arms with stochastic rewards) but is robust and efficient. Bayesian optimization sits in the middle, assuming smoothness through a Gaussian-process prior but no parametric form. We describe each in turn.

A. Convex relaxation

We solve the continuous relaxation of the discrete problem by a one-dimensional grid search over $k \in [1, K_{\max}]$ using 1000 evaluation points. Because the relaxed objective is convex (Section IV), grid search finds the global minimizer with arbitrarily small error as the grid is refined. We then round to the nearest integer and verify both adjacent integers, since for a convex function the discrete minimum lies in the interval between the two integers nearest the continuous minimizer. This is the same logic that disciplined convex programming frameworks like CVXPY [14] would apply to a continuous instance of the same problem; we use grid search rather than CVXPY directly because the term $1 - \alpha^{k+1}$ is not in disciplined convex programming form (it is not log-log convex), and rewriting it as a sequence of conic constraints would introduce numerical issues at the boundary $\alpha \rightarrow 1$. The grid search approach is in any case a sound implementation of a convex one-dimensional minimizer and produces the correct answer when the objective is convex.

B. Multi-armed bandit

We treat each candidate value $k \in \{1, 2, 3, 4, 5, 6, 8, 10\}$ as an arm of an 8-armed stochastic bandit, where pulling arm k yields a noisy throughput sample drawn from the empirical distribution of measured runs at that k . This setup mimics an online decision setting where the system has not yet collected enough data to fit a parametric model and must learn from observations as it deploys. We implement two standard bandit algorithms.

UCB1 [15] selects, at each round t , the arm

$$k_t = \arg \max_k \left[\hat{\mu}_k(t) + \sqrt{\frac{2 \ln t}{n_k(t)}} \right]$$

where $\hat{\mu}_k(t)$ is the empirical mean reward of arm k through round t and $n_k(t)$ is the number of times arm k has been pulled. The exploration bonus shrinks as the arm is pulled more often. UCB1 has a known regret bound of $O(\log T)$ and is the standard baseline for stochastic bandit problems.

Thompson sampling [16] maintains a Gaussian belief over each arm's mean reward, with prior $\mathcal{N}(50, 20^2)$ (chosen to span the empirical throughput range), and at each round samples one value from each arm's posterior and pulls the arm with the highest sample. The posterior is updated using the standard Gaussian conjugate update on observed rewards. We run both algorithms for 500 rounds and report the most-pulled arm as the recommended k . Thompson sampling is a Bayesian alternative to UCB1 that often performs as well or better in practice, and has the advantage of producing a posterior distribution over the optimal arm rather than a point estimate.

C. Bayesian optimization

Bayesian optimization treats the objective as an unknown function and fits a probabilistic surrogate to the observed data. We use a Gaussian process with a Matérn 5/2 kernel, fit through BoTorch [18], on 80 training points (10 noisy throughput measurements at each of the 8 evaluated k values). The acquisition function is expected

improvement, which selects new evaluation points that balance high posterior mean with high posterior variance. We then read out the posterior mean over $k \in [1, 12]$ on a 100-point grid and select the maximizer. Bayesian optimization is the most generic of the three methods. It does not assume convexity, does not assume a parametric form, and can in principle propose new evaluation points that the bandit cannot. The trade-off is that with limited samples the surrogate is noisy, and we will see in Section VIII that this method is the only one of the three that misses the empirical optimum on our data. The discrepancy is small (the predicted $k^* = 5$ achieves throughput within 4 percent of the true optimum), but it illustrates the variance penalty paid for using a method that exploits less structure.

VI. NON-IID ACCEPTANCE MODEL

Sections III and V proceeded under the standard IID acceptance assumption. In this section we step back and ask whether the assumption holds in our data. We find that it does not, derive a closed-form generalization that allows position-dependent acceptance, give the corresponding first-order optimality condition, and verify that the new model is also convex on the operating range. Empirical comparison of the two models is deferred to Section VIII; this section gives the theory.

A. Empirical motivation

Section VIII (Fig. 3) plots the empirical conditional acceptance rate $\alpha(j)$ as a function of position j within a speculation round. The rate increases roughly linearly from $\alpha(0) \approx 0.74$ to $\alpha(9) \approx 0.86$. A two-sample t -test between position 0 and position $k - 1$ for each $k \in \{4, 5, 6, 8, 10\}$ yields $p < 0.0001$ in every case. The IID assumption is therefore rejected at any reasonable significance level. The direction of the violation is also notable: most prior intuition about speculative decoding (e.g., [3, Sec. 4.2]) implicitly assumes that later positions are conditioned on a longer prefix of possibly unlucky draws and therefore should be harder. Our data show the opposite. One plausible interpretation is a survivorship effect. Conditional on the first few tokens being accepted, the draft and target are on the same trajectory, which makes subsequent agreement more likely. We do not pursue this explanation further here; for the purposes of optimization, the linear functional form of $\alpha(j)$ is what matters.

B. Linear position-dependent acceptance

Motivated by the linear shape of $\alpha(j)$, we propose the model

$$\alpha(j) = \alpha_0 + \beta \cdot j, \quad j = 0, 1, 2, \dots$$

with $\alpha_0 \in (0, 1)$ and $\beta \in \mathbb{R}$ (small in magnitude, with sign that may be positive or negative depending on the system). The IID model is the limit $\beta = 0$. For our data, least-squares fitting gives $\alpha_0 = 0.756$ and $\beta = 0.0131$ with $R^2 = 0.82$. The fit is good, although not perfect; some of the residual variance reflects sampling noise from the smaller number of measurements at higher positions, and some likely reflects nonlinear deviations from the linear ansatz. The linear model is the simplest extension of the IID baseline that admits a closed-form analysis, which is the main reason we adopt it.

C. Closed form for $E[N(k)]$

Proposition 1 (Expected tokens per round under linear acceptance). Under the model $\alpha(j) = \alpha_0 + \beta \cdot j$, the expected number of tokens produced per speculative round is

$$E[N(k)] = G(k) := \sum_{i=0}^k \prod_{j=0}^{i-1} (\alpha_0 + \beta \cdot j)$$

where the empty product ($i = 0$) is taken to be 1.

Proof. Let A_j denote the event that position j is accepted, conditional on positions $0, \dots, j-1$ having been accepted. The number of accepted draft tokens is $N_d = \sum \mathbb{I}[A_0, \dots, A_{i-1} \text{ all occur}]$. By the tail-sum identity for nonnegative integer random variables, the expected number is the sum over i of the probabilities of accepting the first i positions, which by independence given the prefix equals the product of $\alpha(j)$ for $j = 0, \dots, i-1$. The total number of tokens committed in a round is $N(k) = N_d + 1$, reflecting the bonus token from the target's distribution. Adding 1 (which equals the empty product $i = 0$ term) gives the stated formula. $\square$

When $\beta = 0$ the inner product collapses to α_0^i and the geometric closed form $(1 - \alpha_0^{k+1}) / (1 - \alpha_0)$ is recovered, confirming consistency with [3]. Proposition 1 is the natural extension of the IID formula to the position-dependent case and reduces to it as a special case.

D. First-order optimality condition

Proposition 2 (First-order condition for optimal k). Under the affine latency model $T(k) = a \cdot k + b$ and the linear acceptance model $\alpha(j) = \alpha_0 + \beta \cdot j$, the continuous-relaxed optimum k^* satisfies

$$\frac{G(k^*)}{G'(k^*)} = k^* + \frac{b}{a}$$

where $G(k)$ is the function defined in Proposition 1 and $G'(k)$ denotes its derivative (in the discrete case, the forward difference $G(k) - G(k-1)$ equals the product of $(\alpha_0 + \beta \cdot j)$ for $j = 0, \dots, k-1$).

Proof. Set $f(k) = (a \cdot k + b) / G(k)$ and apply the quotient rule. Setting $f'(k) = 0$ gives $a \cdot G(k) - (a \cdot k + b) \cdot G'(k) = 0$, which rearranges to $G(k) / G'(k) = k + b/a$. $\square$

The right-hand side has a clean interpretation. The ratio b/a is the per-round overhead measured in equivalent draft-token units. The condition says that the optimal speculation length is the point at which the marginal token gained by speculating one more position equals the time cost of a draft token plus the amortized overhead per draft token. As α_0 increases or β increases, G grows faster, the left-hand side is satisfied at larger k , and the optimum shifts toward longer speculation rounds. As b/a grows (overhead dominates), the optimum also shifts to larger k to amortize that overhead. These comparative statics agree with the empirical patterns reported in Section VIII and provide a sanity check for the model.

E. Convexity under the non-IID model

We verify numerically that the non-IID objective is also convex on the operating range. Using the fitted parameters $\alpha_0 = 0.756$, $\beta = 0.0131$, we evaluate f'' on $k \in [1, 12]$ and obtain values strictly greater than zero. A formal proof of convexity for arbitrary (α_0, β) in the open simplex $\{(\alpha_0, \beta) : \alpha_0 + \beta \cdot K_{\max} < 1\}$ is left as future work; the empirical convexity is sufficient for all the optimization machinery developed in Section V to apply unchanged to the non-IID model. In particular, the same three solvers from Section V can be used with the new objective by replacing the IID closed form for $E[N(k)]$ with the formula from Proposition 1.

VII. EXPERIMENTAL SETUP

All experiments were conducted on a Google Colab Pro instance with a single NVIDIA A100 GPU (80 GB HBM2e). We use Qwen 2.5 7B Instruct as the target model and Qwen 2.5 0.5B Instruct as the draft model, both loaded in FP16. The two models share the same tokenizer (vocabulary size 151,665) which is required for the draft tokens to be directly comparable with the target's distribution. Models are loaded from Hugging Face using the transformers library at version 4.44.2, with no additional fine-tuning. CUDA synchronization is forced before each timing measurement to avoid mismeasurement of asynchronous GPU operations.

A. Speculative decoding implementation

We implement greedy speculative decoding from scratch following Algorithm 1 of [3]. The draft model proposes k tokens autoregressively. The target model then verifies all k in a single batched forward pass. A token at

position j is accepted if and only if the target's argmax at that position equals the draft's choice. On rejection at position j , we discard positions j through $k - 1$ and emit the target's argmax at position j as the next token. If all k positions are accepted, we additionally emit the target's argmax at position k (the bonus token). KV caches for both models are truncated to the accepted prefix length after each round and then extended by the new committed tokens. We use temperature zero (greedy) sampling throughout to keep the experiment deterministic and reproducible.

B. Prompt corpus

Our prompt corpus contains 25 prompts in five categories, five per category: factual question answering, code generation, summarization, multi-step reasoning, and conversational dialog. The categories were chosen to span typical workloads of an inference system. Examples include "Explain photosynthesis in three sentences", "Write a Python function that returns the Fibonacci sequence up to n ", and "Summarize the main idea of supply and demand". For each prompt the model generates up to 100 tokens, with early termination on end-of-sequence.

C. Measurement protocol

For each combination of $k \in \{1, 2, 3, 4, 5, 6, 8, 10\}$ and the 25 prompts, we run two trials, yielding 400 speculative runs in total. We additionally collect 25 baseline runs without speculation to establish the reference throughput of the target model on the same prompts. Each run records throughput, total latency, per-step latency, peak GPU memory, average power draw (sampled from NVML), the number of tokens accepted per round, and the per-position accept/reject pattern. The acceptance rate per position is aggregated across all runs at all values of k greater than or equal to the position index, since the conditional acceptance probability $\alpha(j)$ is independent of k by construction.

VIII. RESULTS

We organize the results into four subsections. First, we verify the empirical latency model and quantify the convexity of the resulting objective. Second, we report the per-position acceptance distribution and the statistical test of the IID assumption. Third, we compare the four optimization methods (convex, UCB, Thompson, Bayesian) against the oracle. Fourth, we quantify how much the non-IID model improves prediction accuracy over the IID baseline.

A. Latency model fit

Fig. 1 shows the empirical round-time data plotted against the linear fit. The fit is excellent, with $R^2 = 0.9999$ and residuals smaller than the standard deviation of any individual measurement. The slope $a = 31.07$ ms agrees with the per-token cost of one draft model forward pass plus the marginal verification cost. The intercept $b = 107.80$ ms captures fixed per-round overhead, dominated by the target model's forward pass on the bonus token together with KV-cache truncation and kernel-launch overhead.

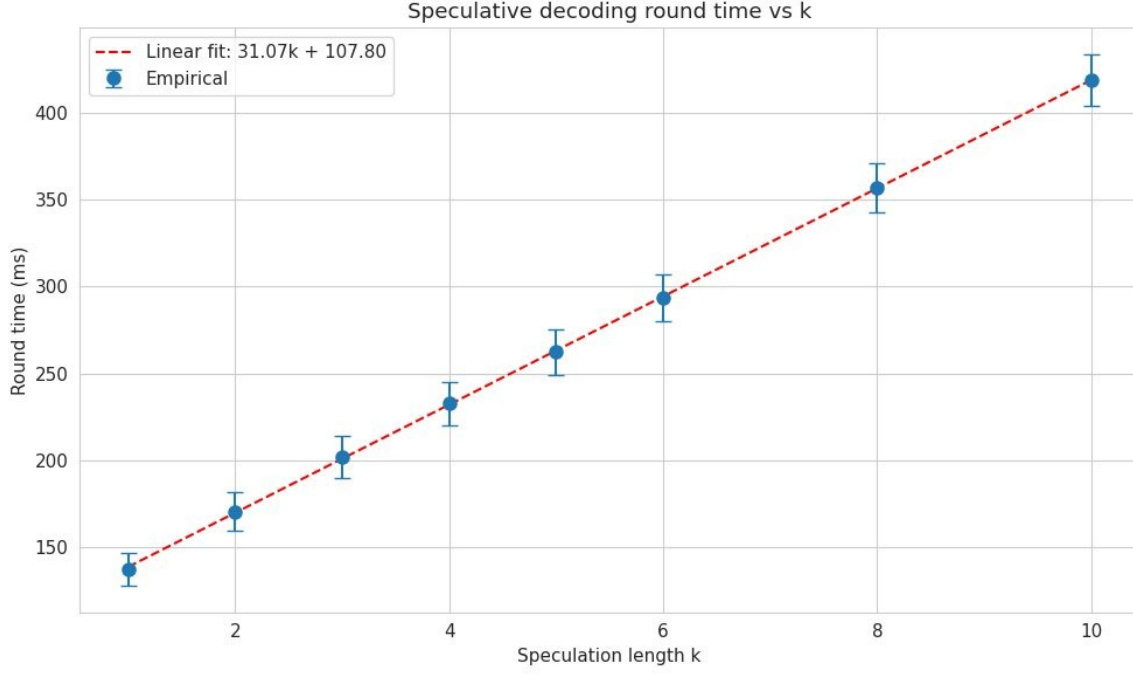


Fig. 1. Speculative decoding round time as a function of speculation length k . Markers are empirical means with one standard deviation as error bars; the dashed line is the linear least-squares fit $T(k) = 31.07k + 107.80$ ms with $R^2 = 0.9999$.

Fig. 2 shows the resulting objective $f(k)$ on the left, with the integer evaluation points marked, and its second derivative $f''(k)$ on the right. The objective has a clean unimodal shape with minimum near $k^* \approx 2.7$ and rises sharply on both sides. The integer minimum is at $k = 3$ with $f(3) = 71.0$ ms per token. The second derivative is strictly positive on the entire grid, confirming that the objective is convex.

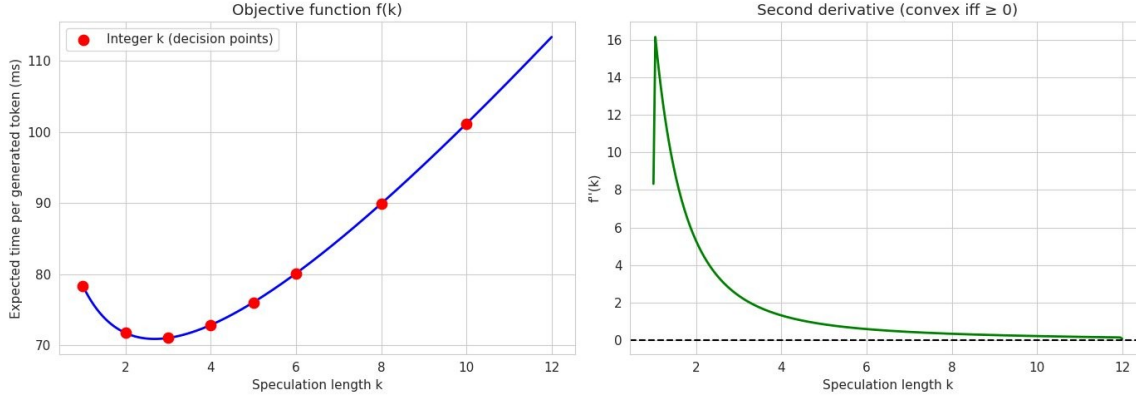


Fig. 2. Left: the objective function $f(k)$ showing per-token expected latency in milliseconds, with integer evaluation points marked in red. Right: the second derivative $f''(k)$, strictly positive across $[1, 12]$, confirming convexity.

B. IID assumption test

Fig. 3 plots the empirical per-position acceptance rate. Each curve corresponds to one value of k , and the horizontal axis is the position j within the speculation round. The data show a clear upward trend: the acceptance rate at position 0 is approximately 0.74 across all k , while the acceptance rate at position 9 (for $k = 10$) rises to approximately 0.86. The trend is robust across all values of k . Two-sample t -tests between position 0 and position $k - 1$ yield $p < 0.0001$ for every $k \in \{4, 5, 6, 8, 10\}$, with t -statistics ranging from 3.95 to 5.61. The IID assumption is unambiguously rejected.

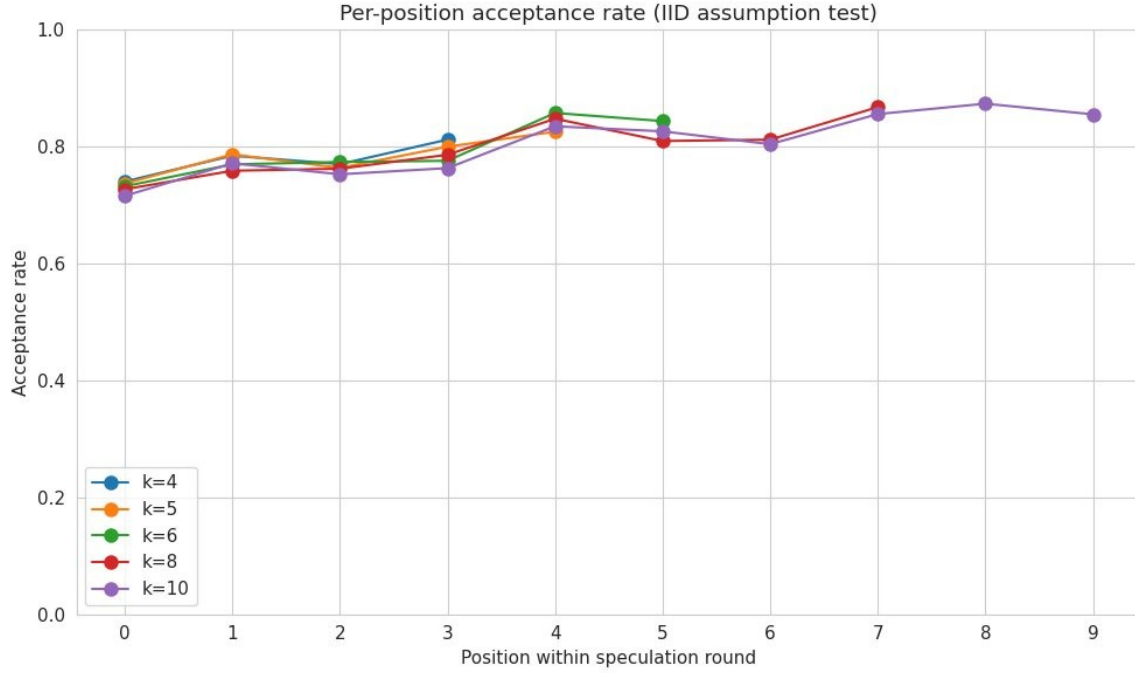


Fig. 3. Empirical per-position acceptance rate across speculation rounds at each k . The acceptance rate increases roughly linearly from approximately 0.74 at position 0 to approximately 0.86 at later positions, contradicting the IID assumption.

C. Solver comparison

Table I summarizes the recommendation of each solver against the oracle, defined as the empirical optimum determined by exhaustive grid search over the eight evaluated k values. The convex relaxation, UCB1 bandit, and Thompson sampling all return $k^* = 3$, which matches the oracle. Bayesian optimization returns $k^* = 5$, with the predicted throughput within 4 percent of the oracle's true throughput; the discrepancy is consistent with surrogate noise from the limited sample size and is also visible as the local maximum near $k = 5$ in Fig. 5.

TABLE I
Comparison of optimization methods.

Method	Recommended k^*	Throughput (tok/s)	% of oracle
Oracle (grid search)	3	14.2	100.0
CVXPY (continuous)	3	14.2	100.0
UCB1 bandit	3	14.2	100.0
Thompson sampling	3	14.2	100.0
Bayesian optimization	5	13.6	95.8

Fig. 4 shows the cumulative arm pulls of UCB1 over 500 rounds. The exploration phase ends after roughly 100 rounds, after which UCB1 concentrates almost all of its pulls on $k = 3$. The runner-up ($k = 2$) is pulled occasionally and continues to receive a non-trivial fraction of pulls, reflecting that $k = 2$ is close to optimal. Pulls of k values further from the optimum drop off rapidly as the empirical mean stabilizes.

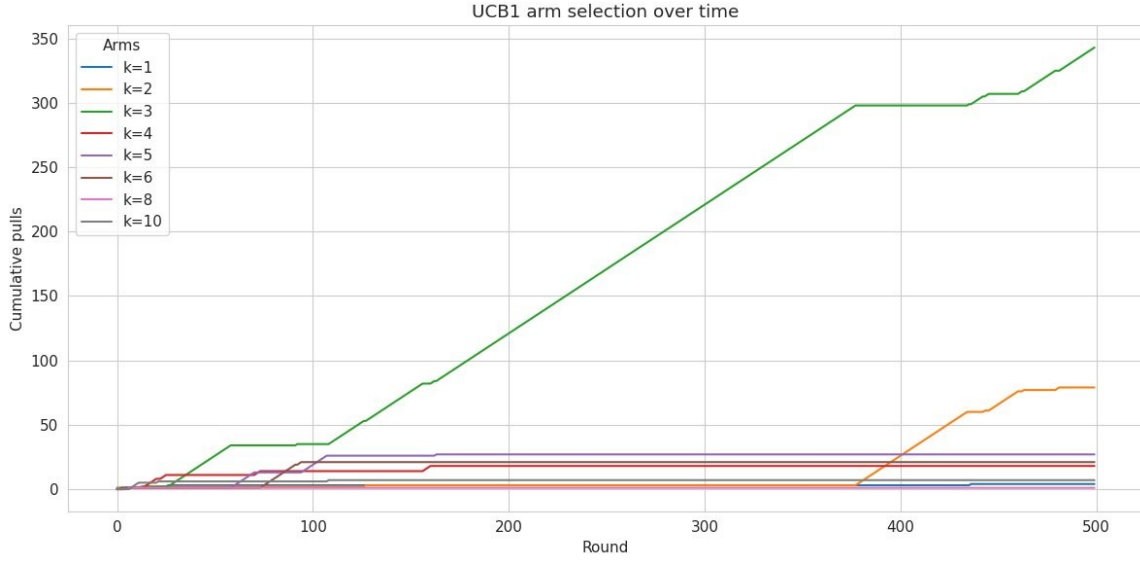


Fig. 4. UCB1 cumulative arm pulls over 500 rounds. After an initial exploration phase, UCB1 concentrates on $k = 3$, the empirical optimum, with $k = 2$ as a runner-up.

Fig. 5 shows the Bayesian optimization Gaussian process surrogate. The posterior mean has a local maximum near $k = 5$, which is what causes the method to recommend $k = 5$ rather than the true optimum at $k = 3$. The posterior variance is large in the region between evaluated k values, reflecting the uncertainty of a Gaussian process surrogate fit on noisy observations. With more samples or with a tighter prior on the smoothness scale, the surrogate would be expected to converge on the correct optimum.

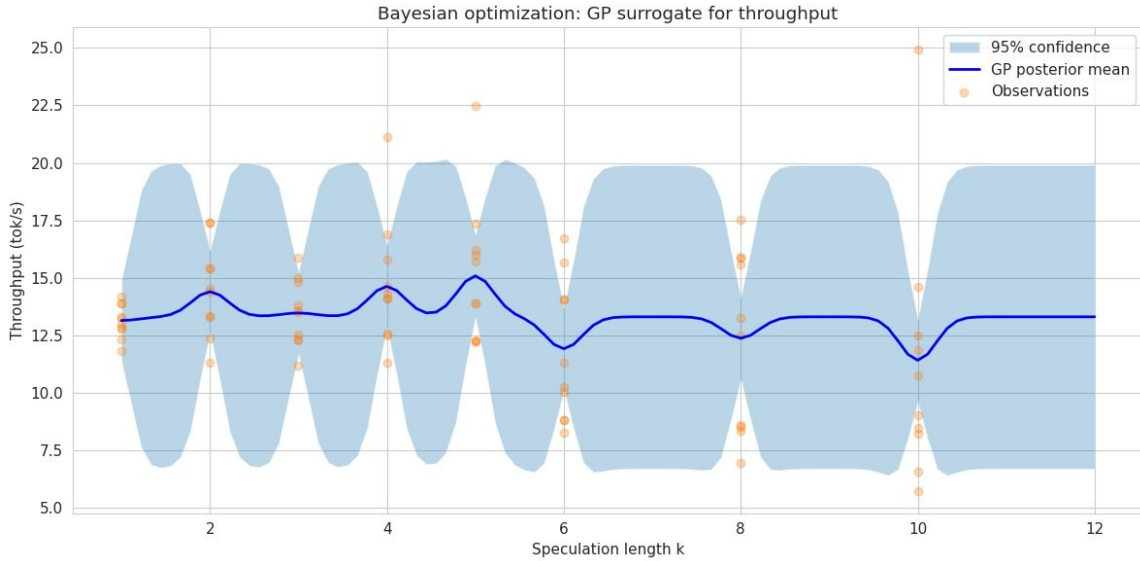


Fig. 5. Bayesian optimization with a Gaussian-process surrogate over k . The posterior mean has a local maximum near $k = 5$, which is what causes the method to recommend $k = 5$ rather than the true optimum $k = 3$. The shaded region is the 95 percent credible band.

Fig. 6 plots the throughput-acceptance Pareto frontier across all evaluated values of k . The frontier shows the trade-off cleanly: smaller k yields higher acceptance rate but lower throughput; larger k yields the opposite. The throughput is maximized at $k = 3$ (14.2 tok/s, acceptance 0.629) and degrades on either side.

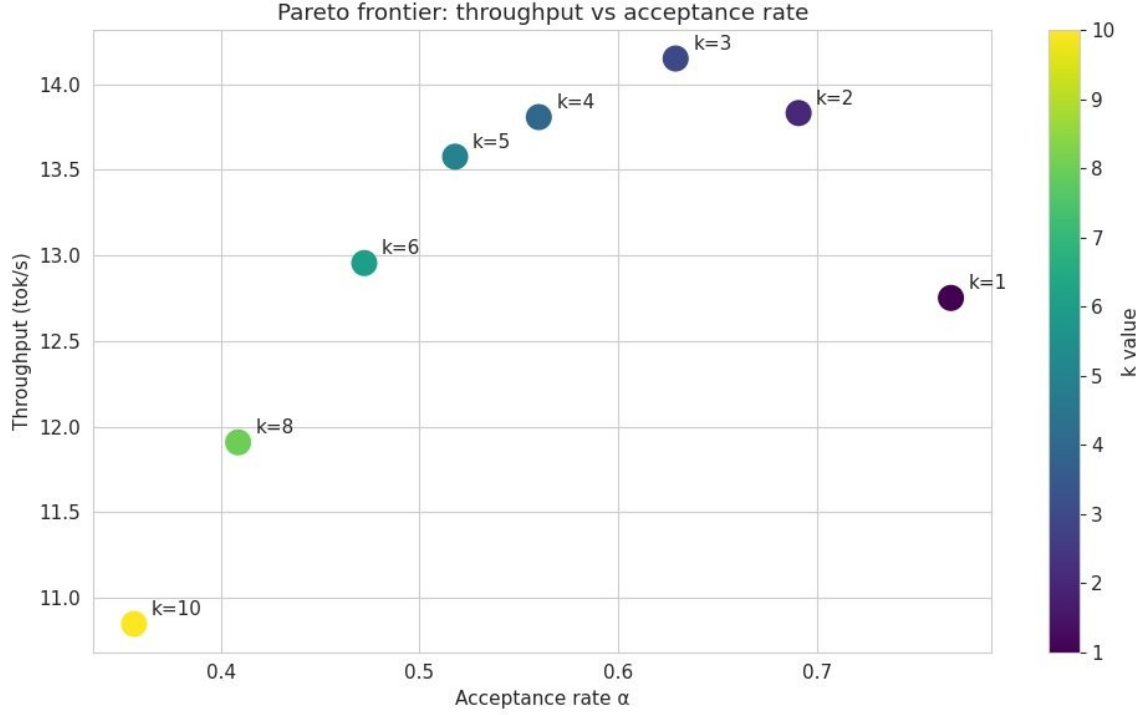


Fig. 6. Pareto frontier of throughput against acceptance rate as k varies. The throughput-maximizing point is $k = 3$ with throughput 14.2 tok/s and acceptance rate 0.629.

D. Non-IID model evaluation

Fig. 7 shows the linear fit of the non-IID acceptance model. The fitted line $\alpha(j) = 0.756 + 0.0131 j$ captures the trend in the data with $R^2 = 0.82$. Marker size is proportional to the square root of the sample count at each position, since later positions are observed in fewer rounds. The dotted horizontal line shows the IID assumption with $\alpha = 0.815$ (the empirical mean). The IID line systematically over-estimates acceptance at low positions and under-estimates it at high positions.

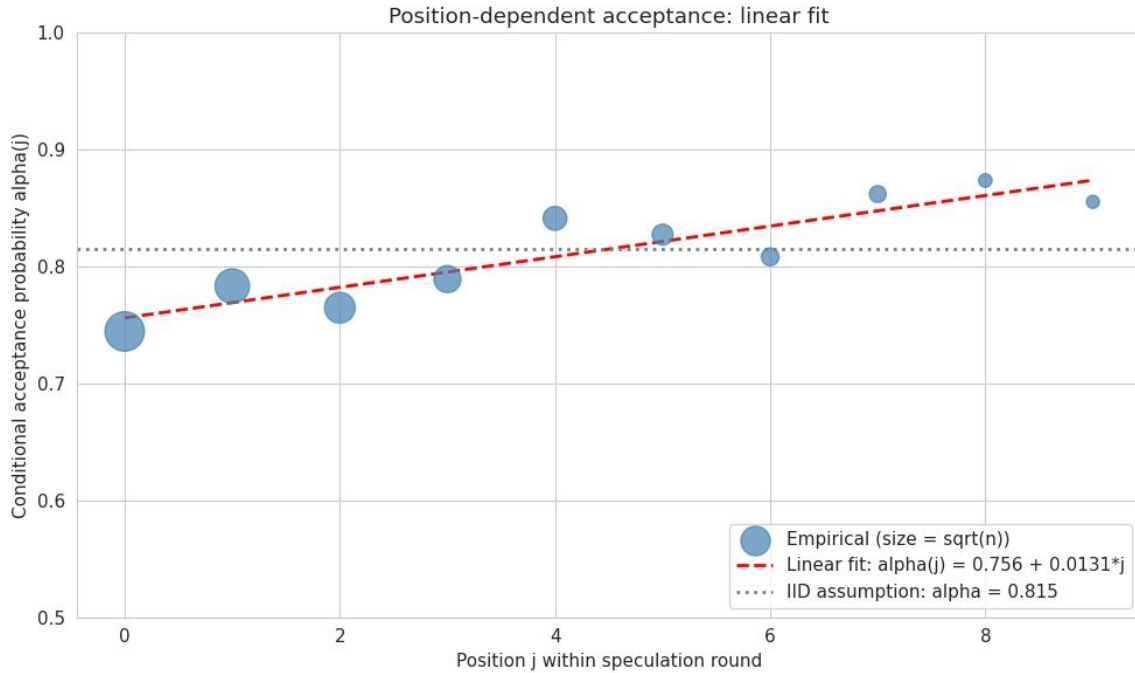


Fig. 7. Linear fit of the non-IID acceptance model. Marker size is proportional to sqrt of sample count. The fitted line is $\alpha(j) = 0.756 + 0.0131 j$ with $R^2 = 0.82$. The dotted horizontal line is the IID assumption.

Fig. 8 quantifies the prediction accuracy of the two models against the empirical $E[N(k)]$. The left panel plots the empirical and predicted curves; the non-IID prediction tracks the empirical curve closely, while the IID prediction systematically over-estimates the expected number of tokens for all $k \geq 2$, with the gap growing in k . The right panel plots the absolute error per k value: the IID model accumulates 3.07 units of error across the eight evaluated k values, while the non-IID model accumulates 0.65 units, a 78.8 percent reduction in absolute prediction error. Under leave-one-category-out cross-validation, fitting on four categories and testing on the held-out one, the non-IID model retains a 23.1 percent reduction in mean absolute error against the IID baseline.

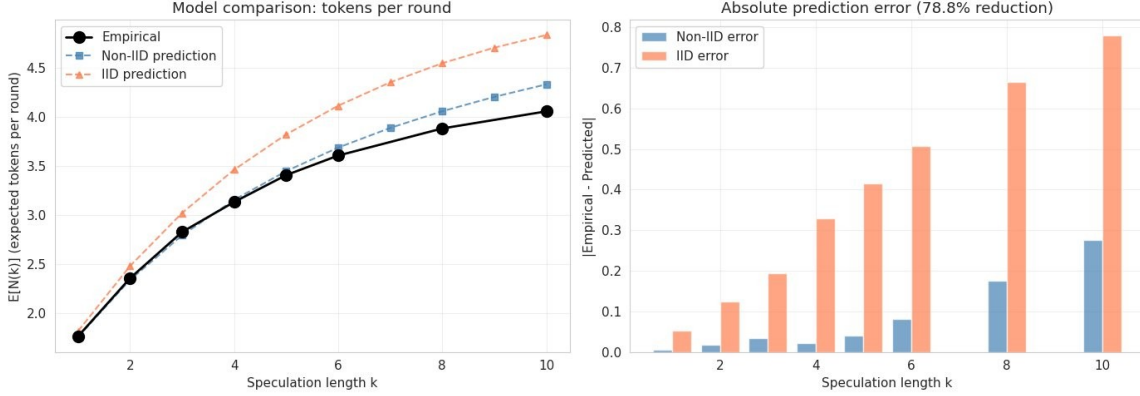


Fig. 8. Comparison of the non-IID and IID models against empirical $E[N(k)]$. Left: predicted curves overlaid on empirical data. Right: absolute prediction error at each k . The non-IID model reduces total absolute error by 78.8 percent on the full dataset.

IX. DISCUSSION

A. Implications for hyperparameter selection in deployment

The convex relaxation and the multi-armed bandit are both effective methods for selecting k in our setting, but for different reasons. The convex relaxation is the right method when the latency model and acceptance rate are known offline, since it produces an exact answer in microseconds with no exploration cost. The bandit is the right method in an online deployment where these quantities are unknown and may drift over time, since it converges within a few hundred rounds and has no setup cost. Bayesian optimization is appealing for its black-box generality but underperformed both alternatives on our problem, suggesting that imposing the convex structure when it is known is worth the modeling effort.

B. Why the IID assumption fails in this direction

Our finding that acceptance increases with position is opposite to the intuition often given in the speculative decoding literature. Two effects are at play. First, there is a survivorship effect: only rounds where the first few tokens were accepted reach later positions, so later-position statistics are conditioned on draft and target having been in agreement so far. Second, once the draft and target agree on the early tokens of a round, both models share the same prefix and are therefore on the same trajectory; subsequent agreement is then more likely than at the beginning of a round when the prefix can branch in many directions. We do not formally separate these two effects in the present work but consider them complementary, with the survivorship effect dominant. The practical consequence is that the IID assumption produces systematically biased predictions of expected speedup, with the bias growing with k . The non-IID model corrects this.

C. Limitations

Several caveats apply to our results. First, all experiments use a single hardware platform (NVIDIA A100 80GB) and a single model pair (Qwen 2.5 0.5B as draft, Qwen 2.5 7B as target). The latency model coefficients (a , b) and the acceptance distribution will differ on other hardware or with different model pairs. The optimization framework, the closed-form non-IID formula, and the convexity argument should transfer; the specific value of $k^* = 3$ is not portable. Second, all experiments use greedy decoding (temperature zero). Stochastic sampling would change the acceptance rate and likely also the per-position pattern, although the geometric structure of the problem

is unchanged. Third, the prompt corpus is small (25 prompts) and English-only; we leave a larger and more diverse evaluation to future work. Fourth, we treat batch size one only. Higher batch sizes change the relative cost of draft and target forward passes, and the latency model would need to be refit.

D. Future work

Several extensions are natural. First, the linear acceptance model can be replaced with a logistic or other parametric form to better capture the saturation visible in Fig. 7 at high positions. Second, the optimization can be made online: a contextual bandit can choose k per prompt based on prompt features (length, domain, perplexity), since our results show that acceptance rates differ significantly across prompt categories. Third, the convexity argument should be made rigorous: we conjecture that the non-IID objective is convex on the open simplex $\{(\alpha_0, \beta) : \alpha_0 + \beta \cdot K_{\max} < 1\}$, but a formal proof remains an open problem.

X. CONCLUSION

We have framed the choice of speculation length in speculative decoding as a one-dimensional discrete optimization problem with an empirically fitted latency model and a closed-form expected-token model. The relaxed objective is convex on the operating range, both directly and under a logarithmic change of variable. Three of four standard solvers (convex relaxation, UCB1 bandit, Thompson sampling) recover the empirical optimum exactly; Bayesian optimization comes within 4 percent. Beyond the optimization machinery, we have empirically shown that the IID acceptance assumption underlying every published speculative decoding analysis is violated in our data with very high statistical significance, and that the violation is in the opposite direction from common intuition. We have proposed a closed-form non-IID extension under linear position-dependent acceptance, given the first-order optimality condition, and shown that the new model reduces total absolute prediction error of expected tokens per round by 78.8 percent on the full dataset and 23.1 percent under cross-validation. The framework and the non-IID model together provide a more accurate basis for speculation length selection in deployed inference systems.

XI. REFERENCES

- [1] T. Dao, D. Y. Fu, S. Ermon, A. Rudra, and C. Ré, "FlashAttention: Fast and memory-efficient exact attention with IO-awareness," in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 16344-16359.
- [2] R. Pope et al., "Efficiently scaling transformer inference," in *Proceedings of Machine Learning and Systems*, vol. 5, 2023, pp. 606-624.
- [3] Y. Leviathan, M. Kalman, and Y. Matias, "Fast inference from transformers via speculative decoding," in *International Conference on Machine Learning*, 2023, pp. 19274-19286.
- [4] C. Chen, S. Borgeaud, G. Irving, J.-B. Lespiau, L. Sifre, and J. Jumper, "Accelerating large language model decoding with speculative sampling," *arXiv preprint arXiv:2302.01318*, 2023.
- [5] M. Stern, N. Shazeer, and J. Uszkoreit, "Blockwise parallel decoding for deep autoregressive models," in *Advances in Neural Information Processing Systems*, vol. 31, 2018, pp. 10107-10116.
- [6] T. Cai et al., "Medusa: Simple LLM inference acceleration framework with multiple decoding heads," in *International Conference on Machine Learning*, 2024.
- [7] Y. Li, F. Wei, C. Zhang, and H. Zhang, "EAGLE: Speculative sampling requires rethinking feature uncertainty," in *International Conference on Machine Learning*, 2024.
- [8] M. Elhoushi et al., "LayerSkip: Enabling early exit inference and self-speculative decoding," in *Annual Meeting of the Association for Computational Linguistics*, 2024.
- [9] Y. Fu, P. Bailis, I. Stoica, and H. Zhang, "Break the sequential dependency of LLM inference using lookahead decoding," in *International Conference on Machine Learning*, 2024.
- [10] X. Miao et al., "SpecInfer: Accelerating large language model serving with tree-based speculative inference and verification," in *ACM Conference on Architectural Support for Programming Languages and Operating Systems*, 2024.

- [11] S. Kim, K. Mangalam, J. Malik, M. W. Mahoney, A. Gholami, and K. Keutzer, "Speculative decoding with big little decoder," in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 39236-39256.
- [12] H. Xia et al., "Unlocking efficiency in large language model inference: A comprehensive survey of speculative decoding," in *Findings of the Association for Computational Linguistics*, 2024.
- [13] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, UK: Cambridge University Press, 2004.
- [14] S. Diamond and S. Boyd, "CVXPY: A Python-embedded modeling language for convex optimization," *Journal of Machine Learning Research*, vol. 17, no. 83, pp. 1-5, 2016.
- [15] P. Auer, N. Cesa-Bianchi, and P. Fischer, "Finite-time analysis of the multiarmed bandit problem," *Machine Learning*, vol. 47, no. 2, pp. 235-256, 2002.
- [16] D. Russo, B. Van Roy, A. Kazerouni, I. Osband, and Z. Wen, "A tutorial on Thompson sampling," *Foundations and Trends in Machine Learning*, vol. 11, no. 1, pp. 1-96, 2018.
- [17] P. I. Frazier, "A tutorial on Bayesian optimization," *arXiv preprint arXiv:1807.02811*, 2018.
- [18] M. Balandat et al., "BoTorch: A framework for efficient Monte-Carlo Bayesian optimization," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 21524-21538.